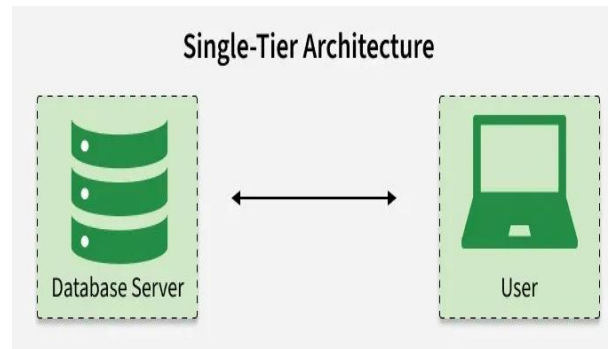


1 tire 2 tire 3 tire architecture

1-Tier Architecture

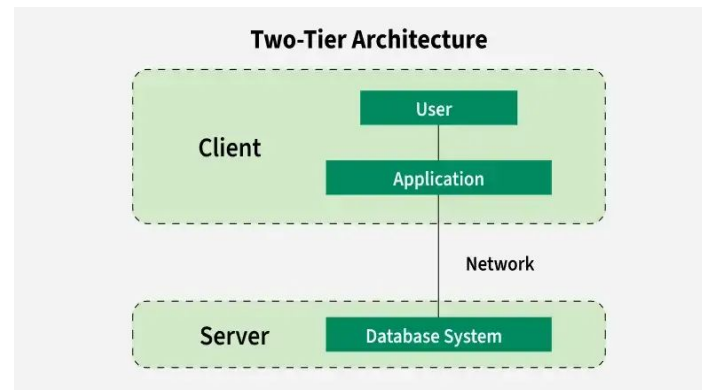
- 1-Tier Architecture: the user works directly with the database on the same system. This means the client, server, and database are all in one application. The user can open the application, interact with the data, and perform tasks without needing a separate server or network connection.



- **Advantages**
- **Simple Architecture:** Only a single machine is required to maintain it.
- **Cost-Effective:** No additional hardware is required for implementing 1-Tier Architecture.
- **Easy to Implement:** It is mostly used in small projects.
- **Disadvantages**
- **Limited to Single User:** It's not designed for multiple users or teamwork.
- **Poor Security:** Everything is on the same machine, if someone gets access to the system, they can access both the data and the application easily.
- **No Centralized Control:** Data is stored locally, so there's no central database. This makes it hard to manage or back up data across multiple devices.
- **Hard to Share Data:** Everything is stored on one computer.

2tire

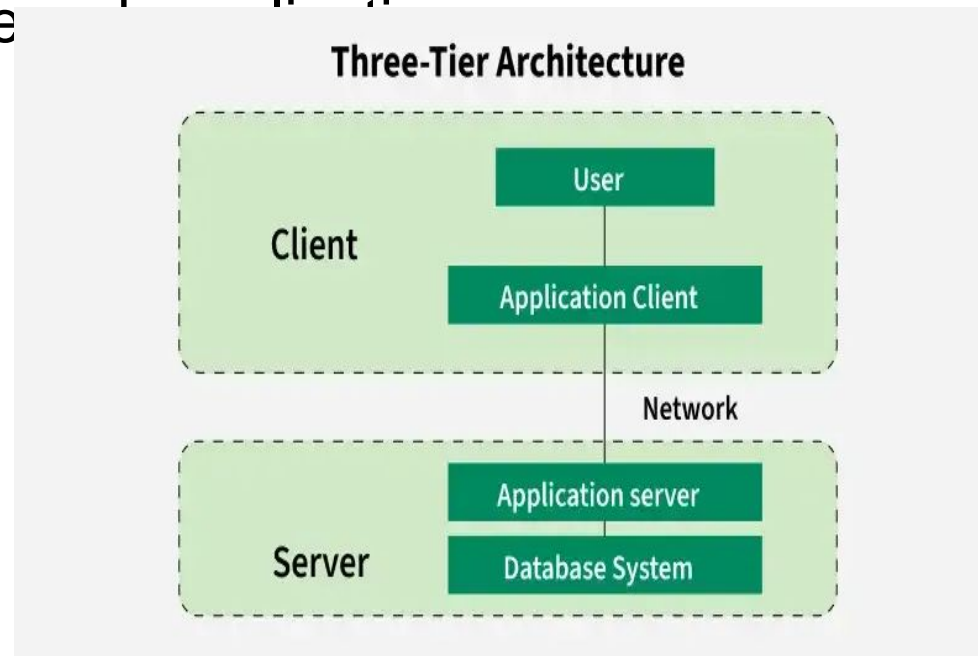
- The 2-tier architecture is similar to a basic client-server model. The application at the client end directly communicates with the database on the server side. APIs like ODBC and JDBC are used for this interaction. The server side is responsible for providing query processing and transaction management functionalities.
- DBMS 2-Tier Architecture On the client side, the user interfaces and application programs are run. The application on the client side establishes a connection with the server side to communicate with the DBMS



- **Advantages**
- **Easy to Access:** Makes fast retrieval.
- **Low Cost:** 2-Tier Architecture is cheaper than 3-Tier Architecture and Multi-Tier Architecture.
- **Easy Deployment:** Easier to deploy than 3-Tier Architecture.
- **Simple:** It consists only two components.
- **Disadvantages**
- **Limited Scalability:** As the number of users increases, the system performance can slow down because the server gets overloaded with too many requests.
- **Security Issues:** Clients connect directly to the database, which can make the system more vulnerable to attacks or data leaks.
- **Tight Coupling:** If the database changes, the client application often needs to be updated too.
- **Difficult Maintenance:** Managing updates, fixing bugs, or adding features becomes harder

3-Tier Architecture

- In 3-Tier Architecture, there is another layer between the client and the server. The client does not directly communicate with the server. Instead, it interacts with an application server which further communicates with the database system and then the query processing and transaction management takes place. This intermediate layer acts as a medium for the exchange of partially processed data between the server and the client. This type of architecture is used in the case of large



- **Example:** E-commerce Store
- **User:** You visit an online store, search for a product and add it to your cart.
- **Processing:** The system checks if the product is in stock, calculates the total price and applies any discounts.
- **Database:** The product details, your cart and order history are stored in the database for future reference.

- **Advantages**
- **Enhanced scalability:** Due to the distributed deployment of application servers. Now, individual connections need not be made between the client and server.
- **Data Integrity:** There is a middle layer between the client and the server, data corruption can be avoided/removed.
- **Security:** This type of model prevents direct interaction of the client with the server thereby reducing access to unauthorized data.
- **Disadvantages**
- **More Complex:** From 2-tier to 3-tier adds one extra communication point.
- **Difficult to Interact:** This sort of interaction to take place due to the presence of middle layers.
- **Slower Response Time:** Since the request passes through an extra layer (application server), it may take more time to get a response compared to 2-Tier systems.
- **Higher Cost:** Setting up and maintaining three separate layers (client, server and database) requires more hardware, software and skilled people. This

Database Architecture

- Database Architecture
- ☐ Also included as Components of database management or database structure.
- ☐ A database system is partitioned into modules that deal with each of the
- responsibilities of the overall system.
- ☐ Functional components of a database system can be broadly divided into,
 - ☐ Storage Manager
 - ☐ Query Processor

